

WORKSHEET - NAPLAN Year 9

Number and Algebra Solving Equations (9)

Teacher: Joanne Rust

Exam Equivalent Time: 10 minutes (based on NAPLAN allocation of ~1.25 min/mark)



Questions

1. ☒ Number and Algebra, NAP-42661

Ben spent twice as much money as Mark.

If they spent a total of \$90, how much did Ben spend?

☐ \$15 ☐ \$30 ☐ \$60 ☐ \$180

2. ☒ Number and Algebra, NAP-13256

Ryan and Oliver are buying goldfish for their fish tank.

Oliver buys 3 times the number of goldfish as Ryan does, plus another 5.

Let m be the number of goldfish that Ryan buys.

Which expression represents the number of goldfish Oliver buys.

☐ $(3 \times m) + 5$ ☐ $3 + (5 \times m)$ ☐ $(m \div 3) - 5$ ☐ $3 - (m \div 5)$

3. ☒ Number and Algebra, NAP-95622

A number is multiplied by itself and then 12 is added.

The answer is 28.

What is the number?

4. ☒ Number and Algebra, NAP-19172

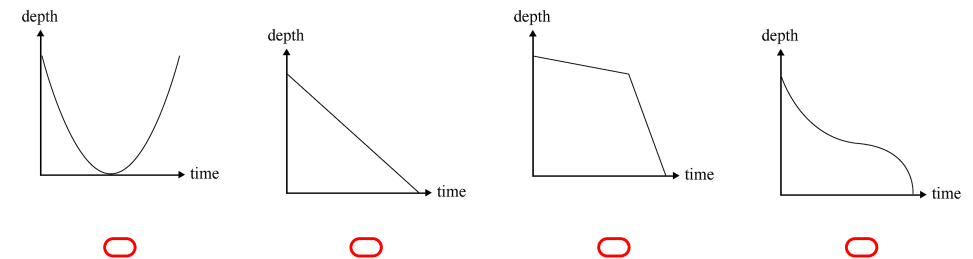
Jack had a full bucket of water.



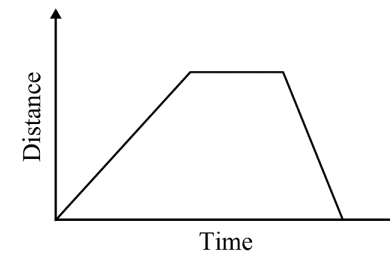
He put a hole in the bottom and the water leaked out.

After a few minutes, Jack put a second identical hole in the bucket and the water poured out more quickly.

Which graph below best shows the depth of water in the bucket against time?



5. ☒ Number and Algebra, NAP-42691



Which of the following situations best fits the distance/time graph above?

- ☐ Matt ran to the park, sat down on a bench, and then walked home.
☐ Matt walked to the park, sat down on a bench, and walked back home.
☐ Matt walked to the park, sat down on a bench and ran home.
☐ Matt walked to the park and back home without stopping.

6. ✕ Number and Algebra, NAP-95681

Dan is a lettuce grower and the cost of fertilising different crop areas is shown in the table below.

size of crop area (m ²)	30	40	50	60	70
cost of fertilising (\$)	2100	2800	3500	4200	4900

Complete the rule that agrees with the values in the table.

cost of fertilising crop = × area of crop

7. ✕ Number and Algebra, NAP-42724

John, Olivia and Louis are picking grapes to earn money.

Their pay is based on the number of tonnes of grapes they pick.

	Tonnes picked
John	4
Olivia	3
Louis	1

Their total pay is \$640.

How much does Olivia earn?

\$

8. ✕ Number and Algebra, NAP-54356

Complete the following equation,

$4^4 - 3^4 + 2^4 - 1^4 =$

Worked Solutions

1. Number and Algebra, NAP-42661

Let $\$x =$ Amount Mark spent

$$2x + x = 90$$

$$x = \$30$$

∴ Ben spent \$60.

2. Number and Algebra, NAP-13256

Goldfish bought by Oliver

$$= (3 \times m) + 5$$

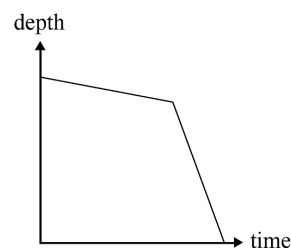
3. Number and Algebra, NAP-95622

$$4 \times 4 + 12 = 28$$

∴ The number is 28.

4. Number and Algebra, NAP-19172

The depth decreases slowly at first and once the second hole is made, it decreases at twice the rate.



5. Number and Algebra, NAP-42691

Matt walked to the park, sat down on a bench and ran home.

STRATEGY: The steep line returning home shows he travelled faster than getting there.

6. Number and Algebra, NAP-95681

$$\begin{aligned}\text{Cost of fertilising} &= \text{cost of 1 m}^2 \times \text{area of crop} \\ &= \frac{2100}{30} \times \text{area of crop} \\ &= 70 \times \text{area of crop}\end{aligned}$$

7. Number and Algebra, NAP-42724

Payment per tonne

$$= \frac{640}{8} = \$80$$

$$\begin{aligned}\therefore \text{Olivia earns} &= 3 \times 80 \\ &= \$240\end{aligned}$$

8. Number and Algebra, NAP-54356

$$\begin{aligned}4^4 - 3^4 + 2^4 - 1^4 \\ &= (4 \times 4 \times 4 \times 4) - (3 \times 3 \times 3 \times 3) + (2 \times 2 \times 2 \times 2) - (1 \times 1 \times 1 \times 1) \\ &= 256 - 81 + 16 - 1 \\ &= 190\end{aligned}$$